It is interesting to note that the peak plasma energy densities observed in this paper for laser pulses focused at NA = 0.8 (U_{avg}≈ 40 kJ/cm^3) are five times higher than the volumetric energy density of TNT (7.54 kJ/cm^3). For NA = 0.3, an even higher peak energy density of 260 kJ/cm^3 was deduced from the photographs in Fig 8. Correspondingly, the fraction of absorbed energy that is converted into shock wave energy is much larger for laser-induced high-density plasmas than for underwater explosions of TNT charges, where $E_{SW}/E_B \approx 1.5$ [58,300].

With growing energy density, the E_{SW}/E_{abs} ratio increases at the cost of the fraction going into vaporization until more than ¾ of the absorbed energy is converted into shock wave energy – but then it does not increase further and also E_{SW}/E_B remains constant. Most of the shock wave energy is dissipated as heat already during the early propagation phase within a few charge or plasma radii, respectively, [42,59,114]. The heat dissipation reduces the amount of mechanical energy available for structural deformation in the plasma vicinity. Moreover, the energy transfer by shock propagation and dissipation at the shock front influences the subsequent cavitation bubble dynamics by lowering surface tension and viscosity [42] and reducing the rate of heat and mass transfer by condensation and heat conduction.

The peak E_{SW}/E_B ratio found in the present study is higher than the value of 2.3 in previous investigations for breakdown at $NA \leq 0.25$ and $U_{avg} \approx 40$ kJ/cm^3 [114]. Experimentally determined E_{SW}/E_B ratios are generally higher than the values obtained in hydrodynamic simulations, in which a homogeneous energy density distribution as starting condition is assumed [42,66]. The neglect of hot spots within the luminescent plasma region leads to an underestimation of the conversion efficiency into shock wave energy, which strongly increases with peak pressure [59,114,300].

Although we have observed that the conversion into shock wave energy increases with average plasma energy density, we are not yet able to assess and quantify the influence of hot spots within the plasma on shock wave generation and energy partitioning. Further insights can be gained by time-resolved photography with ultrahigh spatiotemporal resolution that portrays not only the shock wave emission from the periphery of the luminescent region but also the launching of the shock wave from the hot spot and its propagation through the diffusely luminescent plasma region. Complementary information could be gained through finite volume modeling of hydrodynamics events considering measured absorbed energy and energy distribution in the breakdown volume. Both approaches require further developments of experimental and modeling tools as discussed in section VI.

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5.7. Consequences for laser surgery and material processing in bulk dielectrics

We demonstrated the existence of three basic scenarios for the energy dependence of breakdown effects and enlarges the parameter space in which nanoeffects can be produced. Material processing within the framework of scenarios 1 (fs breakdown) and 2 (IR breakdown at long pulse durations) is already well established. Use of fs breakdown close to threshold allows for precise nanomorphing, cell surgery and dissection [1,13,22,24,27,301-304], while high-density plasmas produced by IR ns pulses enable to create more vigorous laser effects that are, for example, used for membrane disruption in intraocular microsurgery [12]. However, the scientific community has not been aware about the details of scenario 3 for ns breakdown at UV/VIS wavelengths, which imply that nanoeffects can be produced with slm pulses from cost-effective diode pumped Q-switched solid state lasers. Although it has been reported that cell surgery can be performed using visible and UV-A ns laser pulses [124,192,305], it remained a puzzle how nanoeffects could be produced by means of ns pulses as long as the threshold of ns breakdown was identified with plasma luminescence [19]. This question has now been resolved by the discovery and explanation of the two-step behavior of UV and VIS nanosecond breakdown that involves a small-bubble regime without bright plasma luminescence [306,307].

Laser surgery. We demonstrated the usefulness of single-mode UV ns pulses on corneal dissections relevant for refractive surgery in Figs. 12(a) – (c), and first results of animal experiments have been presented by Trost et al. [16]. A potential disadvantage of using UV ns laser pulses for nanosurgery on cells or tissue is the small conversion efficiency of laser energy into mechanical energy close to the bubble threshold (Fig. 9). The applied UV dose is a potential source of photochemical side effects and the mechanical effects associated with plasma-mediated dissection may lower the photochemical damage threshold. However, the photochemical damage potential is at 350 nm four orders of magnitude lower than in the UV-C region below 280 nm [17,308]. Furthermore, the use of vortex beams rather than Gaussian beams for corneal flap dissection diminishes cavitation and shock wave effects, which reduces the aggravation of UV-mediated cell stress by mechanical stress [17,309]. It remains to be investigated under which circumstances photochemical side-effects really counter-indicate the use of 355-nm sub-ns pulses in biophotonics.

Potential applications of UV ns pulses are most advantageous for transparent tissues and cells – for applications in scattering tissues ultrashort laser pulses at IR laser wavelengths remain the best choice because their penetration depth is larger due to a pronounced decrease of scattering with increasing wavelength [188]. Such applications include the induction of skin rejuvenation by subsurface optical

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breakdown in skin [310,311], scar treatment in vocal folds [312] and the attenuation of seizure propagation in the brain by circumscribing cuts around epileptic foci [313]. Linz et al. investigated the required pulse energy for brain interventions as a function of laser wavelength and cutting depth by numerical simulations [37].

Material processing. Generally, UV laser pulses offer a high precision for material processing because of their short wavelength. With ns pulses, nanoeffects can be produced in a broader energy range than with fs pulses, which goes along with a better tuning sensitivity of nonlinear energy deposition (Fig. 5c). The lack of nonlinear propagation effects enables to create filamentation-free localized energy deposition with large volumetric energy densities even at moderate *NA*s. However, this advantage is compromised by the discontinuity of the tuning curve at the transition from the low-density into the high-density plasma regime that may cause problems in some applications.

Figures 12(d) and (e) show microeffects in glass produced in both regimes by 0.56-ns UV laser pulses. With high-density plasmas, tiny cavities and cracks are observed [Fig. 12(d)], similar to those created by tightly focused 100-fs pulses at 780 nm wavelength [194]. The disruptive effects are useful for micro-marking [120]. By contrast, low-density plasmas produce the refractive index changes visible in Fig. 12(e). They may be useful for the writing of optical waveguides [2-4] if glitches into the high-density regime with cavity formation can be avoided. Glitches visible in Fig. 12(e) are mostly located in corners of the scanning pattern, where a larger number of laser pulses were applied. They can possibly be prevented by using higher pulse repetition rates than in the present experiments, which would allow for heat accumulation in the focal region and thermal annealing [314,315]. Another approach could be the use of longer pulses of several ns duration, which feature a broader energy range with nanoeffects than the 0.56-ns pulses used in the present experiment (see Fig. 5c).

The abrupt transition between low- and high-density plasma regimes compromises the reproducibility of laser effects in biological targets featuring inhomogeneous nonlinear absorption properties due to centers of reduced excitation energy for multiphoton ionization. For cells and some cell culture media, a fairly broad transition zone has been observed in which both regimes overlap such that either very small or much larger bubbles are produced [193,242]. Nevertheless, close to the bubble threshold, low-density plasma effects could reliably be create in intestinal epithelium using 355-nm, 560-ps pulses [242]. The discontinuity of the $R_{max}(E_L)$ curve does not matter for applications in homogeneous breakdown media in which the tuning range of interest belongs completely to either the low- or the high-density plasma regime. In some cases, easy switching between both modes may even be desired.

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